Acquiring the first-event law

Local experiments, marked-path uncertainty, and decisions before rare failure

Research continuation with preserved governing sources

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Abstract

We construct a finite interface from local experimental evidence to the joint first-disconnection time, terminal state and a requested recent precursor window in a declared interacting Gibbs reliability model. The abstract uncertainty bridge applies more generally whenever its finite-chain interfaces are supplied. The first route acquires a reusable model and certifies its full event-record law. Its central new bound compares short marked-Palm proposals in relative entropy, controls their accepted submeasures in Hellinger distance, and separately controls the geometric waiting clock. The resulting uncertainty budget has no inverse global-rarity or mean lifetime factor. The second route uses controlled local experiments as probability coins. Balanced-field sampling supplies true static updates; censored release races and a finite Bernoulli factory supply true operating updates without fitting a fine kinetic table. An explicit weight-ratio construction closes the unknown-static-weight interface.

Both routes compose with the completed finite first-event sampler and a paid safe preparation, under a certified safe-mass floor, to give bounded-loss guarantees for a finite menu of constant interventions. They retain their different access and costs: the first pays fine acquisition once, while the second consumes fresh physical queries for later forecasts. Model closure, faithful copies, calibrated fields, priced pins, timing, mixing and static contrast margins remain substantive. The result is a mathematical and experimental-interface construction, not an economical prototype. New executions cover finite primitives and certificate arithmetic, not a general large Gibbs engine or a physical apparatus. Independent foundational and mechanical constructions, all earlier originals and useful failed attempts are preserved in the companion package.

1 The question and the completed consequence

Can local controlled experiments support a forecast of *what fails, when it fails, and the recent sequence leading to it*, without waiting for a global failure? Can that forecast choose an actual permitted intervention with a quantitative loss guarantee?

The completed consequence is conditional but stronger than a CDF or mean certificate. For all-terminal network disconnection under the declared interacting local Gibbs and clock model, the experimental interfaces described below construct a finite generator for the entire requested event record. With specified confidence over acquisition and shared calibration, its output law is within a specified total-variation error of the physical mission law. Every bounded loss of that record is consequently controlled. A finite menu can be compared before deploying the chosen intervention and performing its finite safe preparation. There is no unsuccessful full-lifetime simulation factor.

The two realizations are deliberately distinct. A fitted-model generator can be used offline after its fine parameter acquisition; its robustness is proved in Section 2. An experiment-fed generator uses fresh physical randomness and does not export a finely estimated rate table; its detailed construction is in the separate complete proof and the operational derivation below. Neither result asserts that finite local data discover omitted external mechanisms.

1.1 Why the earlier CDF theorem was insufficient

Consider states a, b_1, b_2, start at a, and let the event be $B = \{b_1, b_2\}$. Set

$$Q(a, b_1) = \lambda r, \quad Q(a, b_2) = \lambda(1 - r), \quad Q(b_1, a) = Q(b_2, a) = 1.$$

This is a full-support reversible chain. The first-event CDF is $1 - e^{-\lambda t}$ for every $r \in (0, 1)$, while its terminal law is $(r, 1 - r)$. Replacing $r = 0.99$ by $r = 0.01$ preserves the entire CDF exactly and changes terminal-state

TV by 0.98. Thus even perfect CDF agreement does not supply the missing marked-law interface. The earlier relative whole-CDF operation remains valuable for very small deadline probabilities and is not replaced by the present absolute-TV guarantee.

1.2 The retained record and the paid information

Let Q be a finite irreducible CTMC, B a nonempty proper event set, π its stationary law, and $p = \pi(B)$. A succinct state need not enumerate the entire state space. Choose a common lazy uniformizer $R \geq 2\max_x[-Q(x,x)]$, a certified normalized additive-symmetrization gap $g > 0$ and a stationary mass floor $\zeta > 0$. Write $P = I + Q/R$. The retained record is

$$\mathcal{R}_{H,s} = \big(\lfloor 2^s R\tau \rfloor,\ X_\tau, \text{state and ordered changes in the last } H \text{ normalized units}\big),$$

with precursor timestamps quantized at the same mesh and history truncated at the mission start. Null uniformization updates are not printed. A physical window h pays $H = \lceil Rh \rceil$. The complete uneventful past is not retained. Its output length would itself carry a lifetime cost.

All TV comparisons concern this same finite record. A finite-valued timestamp is not claimed TV-close to an unrounded continuous time. The accepted finite-bit compiler pays clock digits, update caps, conditional-state error, rejection caps, retained history and memory.

1.3 Precisely what is reused

The first-event sampler, finite clock, acquisition/preparation and correlated static constructions at fixed source revision `719e201332992002528c221c49f80f69431485f6` are accepted machinery [1]. In the rare branch their central identity is

$$\frac{d\mathsf{P}_L}{d\mathcal{L}_\pi(X_{[0,L]})} = \frac{N_B}{p(1+L)}, \qquad w = \frac{\mathbf{1}_{\{H<\tau\leq L\}}}{N_B}. \tag{1}$$

The proposal starts from a rare stationary state placed either at time zero or at a uniformly marked potential event, then evolves forward and backward using P, P^*. Its acceptance mean is $\mu = q_G/[p(1+L)]$. For $L \geq 16(H+1)/g$ and $Lp \leq 1$, $\mu \geq g/8$. This is the nonrare acceptance floor that the new bridge uses. Long waiting is represented by a calibrated geometric integer, not by executing all intervening updates. Buffers handle dependence and arbitrary supplied initial states at their proved structural cost.

The correlated static solver still requires its declared factor graph, certified pinned sampling/mixing and fixed conditional log-surprisal contrast. The new experiment-fed construction implements these static queries from local experimental access; it does not remove the margins. The original code and conclusions are preserved, rather than edited to fit the new theorem.

2 A quantitative bridge from model uncertainty to event records

This section constructs the missing marked-path comparison. It does not deduce it from the older CDF comparison. The two comparisons have different targets and remain distinct.

2.1 Two-model input and the retained record

Let $Q, \widehat{Q}$ be finite irreducible generators on the same encoded state space, with invariant laws $\pi, \widehat{\pi}$ and the same nonempty proper event set B. Use a common rational $R \geq 2\max\{\max_x[-Q(x,x)], \max_x[-\widehat{Q}(x,x)]\}$ and put $P = I + Q/R$, $\widehat{P} = I + \widehat{Q}/R$. Assume both normalized additive symmetrizations have gap at least $g > 0$ and both invariant laws have mass floor $\zeta > 0$. Stationary reverses are denoted by stars. Common transition support is a premise for the finite divergence envelopes below.

The event record has the same meaning as in the accepted first-event construction: the quantized first time, terminal state and ordered changes in the last H normalized time units, truncated at the supplied initial time. The comparison first proves a statement for continuous records; applying any common finite timestamp quantization cannot increase TV.

Define finite bounds

$$\chi_B \geq D_{\mathrm{KL}}(\pi(\cdot \mid B)\|\widehat{\pi}(\cdot \mid B)), \tag{2}$$

$$\kappa_+ \geq \sup_x D_{\mathrm{KL}}(P(x,\cdot)\|\widehat{P}(x,\cdot)), \tag{3}$$

$$\kappa_- \geq \sup_x D_{\mathrm{KL}}(P^*(x,\cdot)\|\widehat{P}^*(x,\cdot)), \tag{4}$$

$$d_p \geq |\log(\pi(B)/\widehat{\pi}(B))|. \tag{5}$$

These are model-confidence bounds, not assumed observations of a rare trajectory. A local experimental construction of them is given below.

2.2 Square-root distance of marked proposals

Write $\mathsf{h}^2(\nu,\omega) = \int(\sqrt{d\nu} - \sqrt{d\omega})^2$ also for subprobability measures. For probabilities, $d_{\mathrm{TV}}(\nu,\omega) \leq \mathsf{h}(\nu,\omega)$ and $\mathsf{h}^2(\nu,\omega) \leq D_{\mathrm{KL}}(\nu\|\omega)$.

Use the accepted length-L marked-Palm proposal: its initial-mark branch has probability $1/(1+L)$ and its internal-mark branch has probability $L/(1+L)$. The two models use the same branch, Poisson count, marked position and uniform timestamps; only their conditional rare state and forward/reverse updates differ.

Lemma 1 (Proposal information bound)**.** *For the two marked proposals* $\mathsf{P}_L, \widehat{\mathsf{P}}_L$*,*

$$D_{\mathrm{KL}}(\mathsf{P}_L\|\widehat{\mathsf{P}}_L) \leq D_L := \chi_B + \frac{L(L+2)}{2(L+1)}(\kappa_+ + \kappa_-). \tag{6}$$

Proof. Retain the branch and marked position as auxiliary variables. The relative entropy chain rule first pays χ_B at the marked state and then the appropriate row divergences for all generated updates. The auxiliary count/time laws are identical. In the internal-mark branch, conditional on $n = M+1$ events, the marked position is uniform in $1,\dots,n$. Its expected backward count is $(n+1)/2$ and its expected forward count is $(n-1)/2$. Since $\mathbb{E}M = L$, weighting the two branches gives expected forward and backward counts each equal to $L(L+2)/(2(L+1))$. Dropping auxiliary variables can only decrease KL. □

Let G mean that the first hit within the block lies in $(H, L]$, and N_B count its potential-event states in B, including time zero and null updates. Both proposals have the same measurable acceptance function $w = \mathbf{1}_G/N_B$. Let their acceptance means be $\mu, \widehat{\mu}$.

Lemma 2 (Accepted-submeasure comparison)**.** *If* $\mu, \widehat{\mu} \geq q_0 > 0$ *and* $z = \sqrt{D_L/q_0}$*, then*

$$d_{\mathrm{TV}}(\mathsf{P}_L(\cdot \mid \text{accept}), \widehat{\mathsf{P}}_L(\cdot \mid \text{accept})) \leq z, \qquad |\log(\mu/\widehat{\mu})| \leq 2z. \tag{7}$$

Proof. Multiplication by $0 \leq w \leq 1$ cannot increase square-root distance. For the accepted submeasures $a = w\mathsf{P}_L$ and $\widehat{a} = w\widehat{\mathsf{P}}_L$,

$$\mathsf{h}^2(a,\widehat{a}) = (\sqrt{\mu} - \sqrt{\widehat{\mu}})^2 + \sqrt{\mu\widehat{\mu}}\,\mathsf{h}^2(a/\mu, \widehat{a}/\widehat{\mu}). \tag{8}$$

The left side is at most D_L. The second term gives the TV bound. For the mass bound, use $|\log\sqrt{\mu} - \log\sqrt{\widehat{\mu}}| \leq |\sqrt{\mu} - \sqrt{\widehat{\mu}}|/\sqrt{q_0}$. □

The acceptance denominator is q_0, not $\pi(B)$. Under the accepted uniform block conditions it can be taken as $g/8$. Because the stationary block success probability is $q_G = p(1+L)\mu$, the two geometric parameters obey

$$|\log(q_G/\widehat{q}_G)| \leq d_p + 2z. \tag{9}$$

The Bernoulli coupling bound therefore gives

$$d_{\mathrm{TV}}(\mathrm{Geom}_0(q_G), \mathrm{Geom}_0(\widehat{q}_G)) \leq \min\{1, e^{d_p+2z} - 1\}. \tag{10}$$

The product of this waiting index and an independent accepted path differs by at most the sum of (7) and (10).

2.3 Complete event-law comparison, including the initial state

For an integer buffer b, let $\eta = e^{-gb}/\zeta < 1$. For each model separately let

$$S_Q = \frac{\eta}{1-\eta} + \frac{1+\eta}{1-\eta}\frac{q_Q(H) + q_Q(b)}{q_{G,Q}} + (1+\eta)q_Q(b), \tag{11}$$

and define $S_{\widehat{Q}}$ analogously. These are the full structural bounds of the accepted block construction, including its joint pre-block buffer charge and unnormalized initial survivor law.

Theorem 1 (Joint first-event model stability)**.** *Suppose both models satisfy the block conditions and have acceptance means at least q_0. For every common initial law ν,*

$$d_{\mathrm{TV}}(\mathcal{L}_{\nu,Q}(\mathcal{R}_{H,s}), \mathcal{L}_{\nu,\widehat{Q}}(\mathcal{R}_{H,s})) \le S_Q + S_{\widehat{Q}} + \sqrt{b\kappa_+/2} + z + \min\{1, e^{d_p+2z} - 1\}. \tag{12}$$

An initial-law TV discrepancy adds its value. There is no inverse stationary-rarity or mean-lifetime factor.

Proof. Compare each true event record with its own ideal block construction. Those comparisons cost S_Q and $S_{\widehat{Q}}$. Couple the observed initial length-b paths. Under Poisson uniformization their relative entropy is at most $b\kappa_+$, since they have the same initial law; Pinsker gives the displayed prefix cost. On a matching early hit both algorithms return the same early record. On matching survival their late outputs consist of the independent geometric index and successful path, shifted by the same integer calendar offset. The preceding two lemmas control that product law. All bounds are uniform in ν. Coupling initial states adds an initial-law error by contraction of a Markov channel, and final quantization cannot increase TV. □

2.4 Explicit envelopes acquired from local factors and rates

Suppose $W, \widehat{W}$ are unnormalized Gibbs weights satisfying

$$|\log \widehat{W}(x) - \log W(x)| \le E \quad \text{for every } x, \qquad |\log(\widehat{Q}(x,y)/Q(x,y))| \le a \quad (x \ne y). \tag{13}$$

For the reversible true and learned local models in the accepted active learner, both reverse kernels equal the corresponding forward kernel.

Lemma 3 (Finite local-to-proposal bound)**.** *Under these reversible premises and the common lazy uniformizer,*

$$d_p = 2E, \qquad \chi_B = 2E^2, \qquad \kappa_+ = \kappa_- = (e^a - 1)^2 \tag{14}$$

are valid conservative bounds. In particular,

$$D_L \le 2E^2 + (L+1)(e^a - 1)^2. \tag{15}$$

Proof. Both the full partition sum and its restriction to B lie within $e^{\pm E}$ of their learned counterparts, giving $d_p \le 2E$. The conditional density ratio on B lies in $[e^{-2E}, e^{2E}]$. For any two probabilities with ratio in $[e^{-c}, e^c]$, convexity of minus the logarithm gives $D_{\mathrm{KL}} \le c\tanh(c/2) \le c^2/2$; apply $c = 2E$.

For a row, put $v = \sum_{y \ne x} \widehat{P}(x,y) \le 1/2$. The off-diagonal chi-square contribution relative to the learned row is at most $(e^a - 1)^2 v$. The diagonal contribution is at most $(e^a - 1)^2 v^2/(1-v) \le (e^a - 1)^2 v$. Their sum is at most $(e^a - 1)^2$. Relative entropy is bounded by chi-square divergence. Reversibility supplies the same reverse bound, and $L(L+2)/(L+1) < L+1$. □

For a nonreversible confidence family the same theorem applies with separately certified reverse-row bounds. Stationary weight ratios and incoming rates must construct them. The reversible active learner does not silently acquire arbitrary driven kinetics.

3 A finite acquired certificate and its intervention meaning

3.1 A common branch, with no hidden rare-event threshold

Fix $0 < \varepsilon \le 1/4$. Conservatively replace a normalized gap certificate by $\min(g, 1)$, and define

$$C = \left\lceil \log_2 \frac{128}{\varepsilon\zeta} \right\rceil, \qquad b = \lceil C/g \rceil, \tag{16}$$

$$L = \left\lceil \frac{1024(H + b + 2)}{g\varepsilon} \right\rceil, \qquad T_{\max} = \lceil 16LC/g \rceil + 1. \tag{17}$$

The two models must use these same buffers, block lengths, clock units and event predicate. Count the learned stationary event probability to a positive interval $[\ell, u]$, and widen it to $[p_-, p_+] = [e^{-2E}\ell, e^{2E}u]$ with outward rounding. Refine Count and require $p_+/p_- \le 2$. Under the energy precision below this is a finite relative-counting task. Both true and learned probabilities then belong to this interval. All counting failures have their own confidence budget; this branch rule is used only on the simultaneous good event.

If $p_+ \le 1/(8L)$, use Theorem 1 with $q_0 = g/8$. The accepted structural estimates give $S_Q, S_{\widehat{Q}} < \varepsilon/32$. If $p_+ > 1/(8L)$, then $p_- > 1/(16L)$. Direct simulation to $T = \lceil C/(gp_-) \rceil \le T_{\max}$ suffices. For either model, the killed-semigroup bound $\Pr_\nu(\tau > T) \le \zeta^{-1/2} e^{-gp_-T}$ is at most $\varepsilon/128$. Coupling the two finite uniformized trajectories gives

$$d_{\mathrm{TV}}(\mathcal{L}_{\nu,Q}(\mathcal{R}_{H,s}), \mathcal{L}_{\nu,\widehat{Q}}(\mathcal{R}_{H,s})) \le \sqrt{T\kappa_+/2} + \varepsilon/64. \tag{18}$$

Thus the direct branch does not reintroduce a numerical $1/p$ cost: its certified horizon is bounded by $T_{\max}$ before execution.

For example, the following sufficient, deliberately loose precisions imply a model-law discrepancy below $\varepsilon/4$ in either branch:

$$E \le \frac{\varepsilon\sqrt{g}}{256}, \qquad a \le \min\left\{1, \frac{\varepsilon}{256}\sqrt{\frac{g}{L+1}}, \frac{\varepsilon}{64\sqrt{T_{\max}}}\right\}. \tag{19}$$

Indeed $e^a - 1 \le 2a$, and consequently $D_L \le 3\varepsilon^2 g/32768$, $z \le \sqrt{3}\,\varepsilon/64 < \varepsilon/32$, $d_p \le \varepsilon/128$ and the observed-prefix term is below $\varepsilon/32$. The geometric term is below $\varepsilon/8$. The direct bound is smaller. Finite upward rounding can use strict slack in these inequalities. These are an existence schedule, not an optimized experimental design and not lower bounds.

3.2 A certificate that can use an actually measured acceptance

The uniform schedule is not compulsory. For a chosen b, L, suppose finite calibration provides an outward interval $\widehat{\mu} \in [\widehat{m}_-, \widehat{m}_+]$ including all numerical and sampling biases. Equation (8) yields

$$\mu \ge m_- := \left(\sqrt{\widehat{m}_-} - \sqrt{D_L}\right)_+^2. \tag{20}$$

If $m_- > 0$, use $q_0 = \min(m_-, \widehat{m}_-)$ in the comparison. For either model its structural bound can be evaluated without knowing p exactly:

$$S \le \frac{\eta}{1-\eta} + \frac{1+\eta}{1-\eta}\frac{H + b + 2}{(1+L)m_{\mathrm{low}}} + (1+\eta)p_+(b+1), \qquad \eta = e^{-gb}/\zeta, \tag{21}$$

where m_{low} is the corresponding acceptance lower bound. This uses $q(t) \le p(1+t)$ and $q_G = p(1+L)\mu$. The calibrated proposal must use the intended G and common window; one cannot insert an unrelated observed acceptance rate. If the square-root gap closes, this certificate reports "unresolved" rather than a positive floor. Its finite arithmetic and statistical errors are explicit inputs, not silently erased.

3.3 Local evidence supplies the uncertainty objects

Use the accepted reversible, pairwise binary Gibbs acquisition model. There are N variables, M declared interaction edges, maximum degree d, a finite menu of K constant calibrated interventions u, and full

observed model closure. In intervention u,

$$W^u(x) = \exp\left\{ \sum_i (h_i + u_i)x_i + \sum_{ij} J_{ij}x_i x_j \right\}, \tag{22}$$

$$Q^u(x, x^i) = r_i^u(x_{\partial i})\sigma\big((1 - 2x_i)(h_i + u_i + \sum_j J_{ij}x_j)\big). \tag{23}$$

The unknown context clocks have supplied or acquired valid numerical bounds $0 < r_* \le r_i^u(s) \le r^*$. The additive-field model, local detailed balance, faithful isolated copies, the absence of hidden memory and the declared graph are premises. Finite observations do not certify model completeness.

The accepted balanced-field learner, with parameter tolerance η_0 and kinetic log-rate tolerance ρ_0, gives simultaneous envelopes

$$E_u = (N + 2M)\eta_0 + N(\xi_u + \tau_u), \qquad a_u = \rho_0 + (1 + 2d)\eta_0 + \xi_u + \tau_u. \tag{24}$$

Here ξ_u is deployment-field calibration error and τ_u is the rational factor/control rounding allowance. Errors in actual timing and detector outputs must be included in that learner's confidence certificate; they are not covered by the displayed field allowance. The original complete acquisition proof and experiment record are preserved unchanged. In particular, a biased calibration clock is not assumed to equal an operating clock.

Interval Dobrushin rows, the pinned conditional-probability contrast condition of the accepted correlated static algorithm, rate bounds and a mass floor construct common g, ζ, R for every model in the retained confidence box. A coarse box may first establish strict margins; subsequent acquisition refines inside that box. A procedure with unresolved margins returns that status. It cannot silently turn a mixing assumption into a learned fact. The static solver is used at its original graph, contrast and pinned-mixing scope, including its numerical inverse margins and degree dependence.

Equations (24) and (19), or the tighter explicit certificate (12), now specify an acquisition precision sufficient for the joint event record. This is the new interface. The previous CDF stability theorem alone supplied none of these marked-path terms.

3.4 The complete resource bill for the fitted-model route

Let $D = N + 2M$ and $C_{\mathrm{kin}} = K \sum_i 2^{\deg(i)}$. Omitting only universal numerical constants, the inherited finite schedules use

$$\text{energy reads} \quad O\left(D\left[\log(2 + B_0) + \eta_0^{-2}\right] \log \frac{D(1 + \log(2 + B_0))}{\delta_E} \right), \tag{25}$$

$$\text{energy dwell} \quad O\left(\frac{D}{r_{\mathrm{cal}}} \left[\log(2 + B_0) + \eta_0^{-2} \log(2/\eta_0)\right] \log \frac{D(1 + \log(2 + B_0))}{\delta_E} \right), \tag{26}$$

$$\text{operating releases} \quad O\big(C_{\mathrm{kin}}\rho_0^{-2} \log(C_{\mathrm{kin}}/\delta_Q)\big), \tag{27}$$

$$\text{capped kinetic dwell} \quad O\left(\frac{C_{\mathrm{kin}}}{r_*\rho_0^2} \log(C_{\mathrm{kin}}/\delta_Q) \log \frac{C_{\mathrm{kin}}(1 + \rho_0^{-2})}{\delta_Q} \right). \tag{28}$$

The original constants, finite censoring and timing construction govern these summaries. Priced pins and neighbor clamps, context setup, bit reads, event detector service, field calibration and finite-slew travel are added at their actual costs. In particular, a pin is not a free state reset and a detector's clock digits are not a physical response time guarantee. The degree term and numerical r_*^{-1} can be large.

Once this acquisition is complete, the learned-model sampler pays the accepted finite static, reverse/forward and clock compilation costs. Its conservative rare-branch calibration uses $O(g^{-1}\varepsilon_{\mathrm{sim}}^{-2} \log(1/\delta_{\mathrm{sim}}))$ length-L Palm proposals; a new draw uses at most $O(g^{-1} \log(1/\varepsilon_{\mathrm{sim}}))$ proposals and a length-b initial segment. Each proposal uses one approximate rare stationary state and $O(L + \log(1/\theta))$ finite updates, at the prescribed per-call tolerance θ. Direct-branch updates are bounded by $T_{\max}$ and the same finite truncation rules. The waiting index is generated in binary by the preserved finite geometric-clock compiler; its digits, not its value, are paid. A physical window of duration h requires $H = \lceil Rh \rceil$. Window output and memory scale accordingly.

All formulas depend on local experimental scales, degree, inverse mixing/static margins, desired accuracy and $\log(1/\zeta)$. There is no $1/\pi(B)$ or mean-first-event simulation factor. There may still be exponential costs in degree, physical contrast, mixing margin or apparatus precision. The earlier thirteen-year serial example is not made economical by this theorem; its coarse CDF calibration is not a joint-record certificate.

3.5 Preparation and decisions that the access really permits

The abstract comparison is uniform over a common supplied initial law. For an actual equilibrium-safe mission, use the accepted finite physical preparation only for menu interventions with the certified bound $\pi^u(A) \geq 1/2$, verified by $p_+^u \leq 1/2$ from the same widened count interval. This safe-mass premise is part of the complete operation, not a consequence of a mixing gap. If this premise is to be certified only from a two-sided statistical interval, a strict safe-mass slack (and its required counting precision) is paid; an unresolved boundary returns that status. In the rare branch the displayed event-mass bound already supplies ample safe slack. Under the chosen constant intervention, let the repairable chain evolve for its certified mixing interval, inspect the state at a known completion time, retain a safe result, and repeat up to a finite cap. This requires the physical process to continue through unsafe states during preparation. It is not an experiment on an irreversibly destroyed system. Its conditional successful law ν_u satisfies

$$e^{-\beta_u} \leq \frac{d\nu_u}{d\pi^u(\cdot \mid A)} \leq e^{\beta_u}, \qquad A = B^c, \tag{29}$$

and its abort probability is at most δ_{prep}. The inherited inspection, field ramp, mixing, retry, and detector costs are paid. For the stated safe stationary mass bound, the finite retry cap is logarithmic in $1/\delta_{\mathrm{prep}}$, without a rare-event rejection cost.

The learned safe initial law has safe mass at least $e^{-2E_u}/2$ and can therefore be sampled with finite capped rejection and prescribed work. More generally a supplied safe-mass floor $s_* > 0$ is possible, but both physical preparation and safe-law sampling must pay $O(s_*^{-1} \log(1/\delta_{\mathrm{prep}}))$ attempts (and the learned density-ratio factor). The present constant-cost preparation claim uses $s_* = 1/2$ throughout. The density-ratio bound gives

$$d_{\mathrm{TV}}(\nu_u, \widehat{\pi}^u(\cdot \mid A)) \leq \tanh(\beta_u/2) + \tanh(E_u). \tag{30}$$

Add the numerical safe-sampling tolerance. Combining preparation, Theorem 1 (or (18)) and the finite model sampler yields a completely specified error

$$\varepsilon_{\mathrm{law},u} = \varepsilon_{\mathrm{model},u} + \varepsilon_{\mathrm{sim},u} + \varepsilon_{\mathrm{init},u} + \tanh(\beta_u/2) + \tanh(E_u). \tag{31}$$

This is a conditional-on-success mission law. For an unconditional operation, output an explicit preparation-abort symbol and charge its probability; assigning it loss at most one adds at most δ_{prep} to risk.

For every menu intervention, choose any bounded loss $\ell_u(\mathcal{R}) \in [0,1]$. It may jointly depend on event time, terminal damage and the retained precursor window. On the single simultaneous acquisition/calibration confidence event, the difference between physical and generated expected loss is at most $\varepsilon_{\mathrm{law},u}$. With

$$n \geq \frac{\log(2K/\delta_{\mathrm{dec}})}{2\sigma^2} \tag{32}$$

fresh generated records per intervention, all empirical risks have additional error at most σ, with probability at least $1 - \delta_{\mathrm{dec}}$. Selecting their minimum gives

$$\mathrm{Risk}(\widehat{u}) - \min_{u \in \mathcal{U}} \mathrm{Risk}(u) \leq 2\left(\max_u \varepsilon_{\mathrm{law},u} + \sigma + \delta_{\mathrm{prep}}\right). \tag{33}$$

Any known intervention cost can be included after scaling total loss into the declared bounded range. The event-law TV certificate covers all bounded record losses, but the finite-sample simultaneous statement above is for the declared K evaluated losses; selecting arbitrary losses after inspecting the same records needs its own complexity bound.

The decision is made after acquisition and before deploying the chosen field and starting its preparation. Only the selected system is physically prepared. This does not grant instantaneous feedback on an unheld live network: observing a state, computing a choice and ramping the apparatus take real time. A mission that must remain active during acquisition or decision requires a separately priced holding, shutdown or exposure model. The theorem does not supply such access.

Absolute-TV risk guarantees do not give relative accuracy for every rare damage subtype, nor control unbounded lifetime costs. The acquired relative whole-CDF theorem remains a separate operation and can be used alongside the present joint-record guarantee.

4 A second realization: use the experiments as update coins

The same acquired-event endpoint has a different implementation. A fine kinetic table can be avoided if new local experiments remain available during compilation and every forecast. This changes the experimental

bill; it does not turn a finite data set into an unlimited exact simulator. The following construction was developed independently of the marked-law bridge. Its complete standalone source and execution record are preserved in `next5_decision`.

Use the same binary pairwise model, with known neighborhoods and unknown h_i, J_{ij}, and the actual menu-dependent rates

$$Q^u(x, x^{(i)}) = r_i^u(s)\sigma((1-2x_i)(\ell_i(s)+u_i)), \quad \ell_i(s) = h_i + \sum_j J_{ij}s_j, \quad 0 < r_* \le r_i^u(s) \le r^*. \tag{34}$$

The same $r_i^u(s)$ applies to both values of the focal bit. That is the declared reversible mechanism. A faithful local cell realizes every requested neighborhood, the full operating environment, and the actual focal clock. A held artificial field z has stationary probability $\sigma(\ell_i(s)+z)$ and relaxation rate at least r_{cal}, even though its clock may differ arbitrarily from the operating clock. Coarse acquisition locates a balanced field with probability in $[1/4, 3/4]$ and certifies the required strict static/mixing margins.

Context realization, additive-field calibration, available slew and settling, focal state pins, bit reads, event detection and reference timing have the actual costs

$$C_{\mathrm{context}},\ C_{\mathrm{field}}(\xi),\ C_{\mathrm{slew}},\ C_{\mathrm{settle}},$$
$$C_{\mathrm{pin}},\ C_{\mathrm{read}}(\beta),\ C_{\mathrm{event}}(\beta),\ C_{\mathrm{timer}}(\Delta).$$

Missed exits, dead times, readout error and drift must fit the allocated conditional error budgets. The operation is finite only if those services are supplied at the requested precision. The proof does not build a universal cell, field meter or detector.

For the visible edge-failure bits, coarse intervals must certify

$$0 < q_i^- \le \pi^u(x_i = 1 \mid x_{-i}) \le q_i^+ < 1, \qquad q_i^- \ge (q_i^+)^c, \tag{35}$$

with fixed c, as required by the accepted correlated cut construction. A pinned Dobrushin margin is $\alpha > 0$. If $|\ell_i(s)| \le B_0$, $|u_i| \le U$, one convenient common certificate is

$$b_E = \lceil 2(B_0+U) \rceil, \quad \zeta = 2^{-N(b_E+1)}, \quad \rho = 2r^*, \quad R = N\rho, \quad g = \alpha r_*/R. \tag{36}$$

Every complete state has mass at least ζ: the bound on a full conditional bit probability also bounds partial conditionals by the tower property, and these factor the state probability. Heat-bath Dirichlet-form domination by r_* supplies the normalized gap.

There are two explicit coarse acquisition options. Enumerating all local contexts costs $\sum_i 2^{\deg(i)}$ and is polynomial only at fixed degree. In the pairwise model, the accepted $D = N + 2M$ energy-context learner instead reconstructs all local logits to error $(1+2d)\eta_0$. Choosing $\eta_0 \le 1/[8(1+2d)]$ and field error at most $1/8$ makes $z = -\operatorname{clip}_{[-B_0,B_0]}(\widehat{\ell}_i(s))$ balance every later context without exceeding the declared field range. The sufficient interval row test $\sum_j \tanh((|\widehat{J}_{ij}| + 2\eta_0)/4) < 1$ certifies a pinned Dobrushin margin when it has strict slack. Interval sums of positive and negative couplings bound every conditional logit. This second option avoids scanning an arbitrary clock table, but it requires that the stated sufficient row certificate succeed. Refinement and unresolved-premise outcomes are explicit. Neither option removes the fine field metrology needed in the update construction below.

5 A physical true-Gibbs update without fine energy estimation

Lemma 4 (Balanced two-coin construction)**.** *Suppose a balanced physical coin has success probability $q = \sigma(\ell + z) \in [1/4, 3/4]$. For $a = e^{u-z}$ the following experiment has output law* Bernoulli$(\sigma(\ell+u))$*: choose branch 1 with probability $a/(1+a)$ and branch 0 otherwise; toss a q coin; on branch 1 accept a success and return 1; on branch 0 accept a failure and return 0; otherwise repeat. The acceptance probability on each attempt is at least* $1/4$.

Proof. The probabilities of accepted 1 and accepted 0 are $aq/(1+a)$ and $(1-q)/(1+a)$. Their sum is at least $\min(q, 1-q) \ge 1/4$ and their normalized ratio is $aq/(aq+1-q) = \sigma(\ell+u)$. □

This is a two-coin/Bernoulli-race identity, established machinery [6, 5]; the new use here is its physical interface and composition with rare-event generation.

For a requested $0 < \theta \le 1/4$, set

$$J = 4\lceil \log_2(8/\theta) \rceil, \qquad \beta = \theta/(16J).$$

Cap the experiment after J attempts, returning a fixed fallback bit on exhaustion. The ideal tail is at most $(3/4)^J \le e^{-J/4} < \theta/8$. Before each read hold for $t_{\mathrm{hold}} \ge r_{\mathrm{cal}}^{-1} \log(1/\beta)$. This makes the physical coin within conditional TV β of q. A finite dyadic branch selector has error at most β. The two substitutions over at most J attempts add at most $2J\beta = \theta/8$.

Let the numerical values of each of the actual fields u, z be within $\xi \le \theta/4$. The computed log shift then differs by at most 2ξ; since σ is 1/4-Lipschitz, the resulting infinite-factory output probability differs by at most $\xi/2 \le \theta/8$. Acceptance remains at least $1/4$ for every positive branch weight even with the perturbed shift. Finite exponential/selector evaluation is included in the selector error. Thus the entire capped output has conditional TV at most θ, with slack for the declared response/readout error allocations. Standard interval exponentials and dyadic thresholds require bit cost polynomial in $B_0 + U + \log(J/\theta)$; their cost is not proportional to e^{B_0+U}.

The actual service bill per requested Gibbs update is bounded by

$$\begin{aligned} C_G(\theta) \le\ & C_{\mathrm{context}} + C_{\mathrm{field}}(\theta/4) + C_{\mathrm{slew}} + C_{\mathrm{settle}} \\ & + J\big[r_{\mathrm{cal}}^{-1} \log(16J/\theta) + C_{\mathrm{read}}(\beta) + C_{\mathrm{timer}}(\Delta_G)\big] + C_{\mathrm{arith}}. \end{aligned} \tag{37}$$

The hold-timing resolution Δ_G is chosen so the actual hold is at least the specified time; rounding upwards suffices if its excess time is charged. This method requires no focal-state reset because its bias bound holds uniformly conditionally on every earlier experimental history.

Lemma 5 (Balanced log-odds measurement)**.** *At a balanced context, a finite number of physical readings, $O(h^{-2}\log(1/\beta_0))$, and additive-field calibration to $O(h)$ estimate $\ell_i(s)$ to absolute error h, with conditional failure probability at most β_0. Every reading holds for $O(r_{\mathrm{cal}}^{-1}\log(1/h))$. All costs are uniform over the previous experimental transcript.*

Proof. For $h \le 1$, set $a = h/128$. Hold until each conditional mean differs from the true q_z by at most $a/2$. Martingale Hoeffding on $M \ge 2a^{-2}\log(2/\beta_0)$ observations bounds the empirical error by a, including the conditional-mean bias, with probability at least $1 - \beta_0$. Clip the empirical probability to $[1/8, 7/8]$. Since true $q_z \in [1/4, 3/4]$, clipping cannot increase its error. The logit derivative on the clipping interval is at most $64/7 < 10$, so its error is at most $10a < h/12$. Subtract the measured field z, whose actual value is known to $h/4$, and evaluate the logarithm to $h/4$. The total is less than h. For $h > 1$ one may use the $h = 1$ procedure. The hold and every calibration, response, read and timer service are charged. □

No fine log-odds estimate is used in the Gibbs update of Lemma 4. Lemma 5 is used only when an actual relative static probability needs a local weight-ratio evaluation.

6 True operating updates from censored exits

The actual operating clocks can change with the menu. We never infer them from the clocks at balancing fields. Fix a requested coordinate, its neighborhood, starting bit, and full operating environment u. Pin that bit, release, and let its first-exit time be $T \sim \mathrm{Exp}(\lambda)$, where λ is the applicable rate in (34). It obeys $0 < \lambda \le r^*$.

Lemma 6 (Operating race and finite walk)**.** *Take $\rho = 2r^*$, generate an independent $E \sim \mathrm{Exp}(\rho)$, and censor the focal first-exit observation at E. The bit $\mathbf{1}_{\{T<E\}}$ has probability $p = \lambda/(\lambda+\rho) \le 1/3$. A random walk starting at 0, moving up 1 on a p-coin success and down 1 otherwise, ever hits $+1$ with probability $p/(1-p) = \lambda/\rho$. For $0 < \theta \le 1/4$, put*

$$m = \lceil \log_2(8/\theta) \rceil, \qquad n = 72m.$$

Stop the walk at $+1$, at $-m$, or after n steps; output 1 only if $+1$ has been reached. Its output probability differs from λ/ρ by less than $\theta/4$. If each implemented input race bit has conditional TV error at most $\beta = \theta/(8n)$, the full output error is less than $\theta/2$.

Proof. Independence of the two clocks gives $\mathbb{P}(T < E) = \int_0^\infty \lambda e^{-(\lambda+\rho)t}\,dt$. The infinite-walk hitting probability is the classical biased-walk solution $r = p/(1-p)$, obtained by solving the bounded harmonic hitting equation or optional stopping on finite intervals and taking a limit. Since $r \le 1/2$, the paths lost by absorption at $-m$ have probability at most the eventual success probability starting there, namely $r^{m+1} \le 2^{-m-1} < \theta/16$.

For the unlimited walk, $\mathbb{E}S_n \le -n/3$. A path not yet absorbed at either boundary has $S_n > -m$. Since $n \ge 6m$, Hoeffding for independent increments in $[-1,1]$ gives

$$\begin{aligned}\mathbb{P}(\text{not absorbed at } n) &\le \mathbb{P}(S_n - \mathbb{E}S_n > n/3 - m)\\ &\le \exp[-(n/3-m)^2/(2n)]\\ &\le e^{-n/72} = e^{-m} \le 2^{-m} \le \theta/8.\end{aligned}$$

The reference walk uses independent ideal coins. Conditional kernel errors of the physical implementation are coupled sequentially: the probability any of at most n races differs is at most $n\beta = \theta/8$. Adding the three errors proves the result without assuming the physical coins independent. □

The construction is a particular Bernoulli factory; the general analytic Bernoulli-factory theory is established [7]. The explicit bounded walk, censored physical race and total error/cost accounting are provided because they are needed by the claimed physical operation.

6.1 A finite deadline and comparison device

For each ideal race input at accuracy β, a sufficient finite recipe is

$$D = \rho^{-1}\lceil \log_2(32/\beta)\rceil, \qquad 0 < \Delta \le \beta/(256\rho). \tag{38}$$

The ideal deadline tail is $e^{-\rho D} \le \beta/32$. Use an independent uniform b-bit grid variable to implement the inverse exponential map. Couple it to a true $U \in (0,1)$ by a uniform grid interval. On $E = -\log U/\rho \le D$ the map's derivative is at most $e^{\rho D}/\rho$. A sufficient uniform grid mesh is

$$2^{-b} \le \min\{\beta/256,\ \rho\Delta e^{-\rho D}/32\}.$$

An endpoint cell intersecting $e^{-\rho D}$ has probability at most 2^{-b} and is charged separately. The endpoint $U = 0$ is never evaluated: all grid cells below the cap threshold return the declared fallback. On the other cells, finite interval evaluation and rounding produce a scheduled deadline within Δ of the coupled ideal E. Choosing interval width at most $\Delta/4$ and rounding to a clock grid of spacing at most $\Delta/4$ supplies this bound. The needed bit length is $O(\log(1/\beta) + \log(1/(\rho\Delta)))$; the displayed inequalities can be checked with rational conservative exponential bounds.

Include actual deadline service error and timestamp/response errors in a combined comparison error of at most 3Δ. The physical comparison can then differ from the ideal event only if $|E - T| \le 3\Delta$, or on a cap or numerical failure. Conditional on T, the density of the independent E is at most ρ, so the tie band has probability at most $6\rho\Delta$. Use a fixed rule in ties. Thus cap, threshold cell, rounding and tie-band errors sum to less than $\beta/8$ with the choices above; the remaining budget up to β covers declared read, missed-exit and service errors. If these errors are deterministic accuracy bounds rather than random failures, their contributions are just the tie-band bound. No total variation approximation of a continuous real clock by a discrete variable is asserted; only the event-comparison bit is approximated.

Each race ends by $D + O(\Delta)$, including a no-exit observation. It never waits for an uncontrolled rare full dwell $1/\lambda$. But it uses a new requested-state pin and release before every trial, even when that state is rarely occupied naturally. The pin is a substantive resource.

6.2 Full operating kernel and cost

Choose a coordinate uniformly from N, request its actual operating environment and neighborhood, and use Lemma 6 to flip it with probability λ_i/ρ. The ideal kernel is exactly $P = I + Q^u/(N\rho)$. A finite coordinate selector can use an error budget $\theta/4$; the preceding factory has slack for the remaining budget.

For the actual source and target states, $\pi^u(x)Q^u(x, x^{(i)}) = \pi^u(x^{(i)})Q^u(x^{(i)}, x)$, because the same $r_i^u(s)$ multiplies the complementary logistic probabilities. Therefore $P^* = P$. This route acquires both update

interfaces only for the declared reversible form. It does not construct the reversed kernel of an arbitrary unknown nonreversible process.

A worst-case service cost for an operating update at error θ is

$$\begin{aligned} C_P(\theta) \le{}& C_{\mathrm{context}} + C_{\mathrm{field}}(\xi_{\mathrm{op}}) + C_{\mathrm{slew}} + C_{\mathrm{arith}} \\ &+ n\big[C_{\mathrm{pin}} + C_{\mathrm{settle}} + C_{\mathrm{event}}(\beta) + C_{\mathrm{timer}}(\Delta) + D + O(\Delta)\big], \end{aligned} \tag{39}$$

with m, n, β, D, Δ as above and with any operating-environment reproduction precision ξ_{op} required to produce the declared actual Q^u. When faithful copies reproduce that environment exactly by assumption, $C_{\mathrm{field}}(\xi_{\mathrm{op}})$ denotes that paid reproducible setting. Otherwise the induced kernel difference is separately bounded and allocated to θ; an unknown arbitrary dependence of r_i^u on the field does not permit a continuity inference from field accuracy. Deployment must reproduce the same environment or carry an additional proved law-error bound.

The number of pins is $O(\log(1/\theta))$, and the total censored dwell is $O(\rho^{-1}\log^2(1/\theta))$ with the explicit conservative constants above. Small rates need not be estimated: a small success probability is implemented by a correspondingly unlikely success. This does not improve the cost of *estimating* a small rate relatively. It does not remove the numerical r_*^{-1} dependence of g^{-1} or the cost of physically setting a locally unlikely starting bit.

7 Experimental rare static access without an unknown weight oracle

Lemma 7 (Finite experimental pinned sampler)**.** *For any coordinate pinning D and $0 < \omega \le 1/4$, a conditional-on-history TV-ω draw from $\pi^u(\cdot \mid D)$ uses at most NS Gibbs updates, where*

$$S = \left\lceil \alpha^{-1} \left\lceil \log_2(4N/\omega) \right\rceil \right\rceil, \qquad C_{\mathrm{pinlaw}}(\omega) \le NS\, C_G\big(\omega/(4NS)\big) + C_{\mathrm{virtual}}.$$

The pin constraints here are virtual classical constraints. They are not a claim that the whole physical system can be prepared in the rare pin event.

Proof. Start a virtual configuration in any pin-compatible state. Systematically update its free coordinates, requesting a local cell at the current virtual neighborhood each time. Pinning preserves the influence bounds. For ideal Gibbs updates, if all coordinate disagreement probabilities before a sweep are at most d_0, none exceeds d_0 during it and each updated coordinate has disagreement at most $(1-\alpha)d_0$. A complete sweep gives that bound everywhere. Coupling to a stationary copy gives TV at most $N(1-\alpha)^S \le Ne^{-\alpha S} \le \omega/4$. Substituting conditional-TV-$\omega/(4NS)$ physical updates costs at most $\omega/4$ by sequential coupling. The result has slack for finite control and any charged services. If all coordinates are pinned, return that state. □

The accepted heavy-prefix estimator used the numerical terminal weight $W(x_D)$. It cannot be copied unchanged when W is unknown. The following identity closes precisely that gap.

Lemma 8 (Experimental relative pin mass)**.** *For any nonempty coordinate-pinning event D, rational $0 < v, \beta_0 \le 1/4$, the physical access constructs a positive rational $\widehat{p}_D$ with*

$$\mathbb{P}\big(e^{-v}\pi^u(D) \le \widehat{p}_D \le e^{v}\pi^u(D) \,\big|\, \textit{prior transcript}\big) \ge 1 - \beta_0.$$

Its worst-case number of pinned draws is $O(N^3 v^{-2}\log(N/\beta_0))$; each draw has TV tolerance $O(v/N)$. It additionally uses $O(N^3 v^{-2}\log(N/\beta_0))$ balanced readings to estimate local log odds, at field precision $O(v/N)$ and hold time $O(r_{\mathrm{cal}}^{-1}\log(N/v))$. There is no inverse $\pi^u(D)$ cost.

Proof. Heavy-complete D by visiting its free coordinates. At each stage estimate the current conditional marginal and choose the empirically heavier bit, then add that bit to the prefix. Independently in the algorithmic sense (fresh future experiments, not a physical reset assumption), heavy-complete the empty pin event. Denote their terminal states by x_D, x_0, and the products of their true selected conditional probabilities by H_D, H_0. The exact identities

$$H_D = \frac{\pi^u(x_D)}{\pi^u(D)}, \qquad H_0 = \pi^u(x_0)$$

give

$$\pi^u(D) = \frac{W^u(x_D)}{W^u(x_0)} \frac{H_0}{H_D}. \tag{40}$$

When D pins every coordinate, $H_D = 1$ and the same identity applies.

Put $a = v/(32N)$. Estimate each of at most $2N$ marginal stages using

$$M \geq 2a^{-2} \log(8N/\beta_0)$$

pinned draws, each at conditional TV at most $a/2$. Conditional mean bias is at most $a/2$, and bounded martingale Hoeffding bounds the empirical deviation around its conditional means by $a/2$ with failure probability at most $\beta_0/(4N)$. A union bound gives a simultaneous event of probability at least $1 - \beta_0/2$. On it, every empirically selected true heavy probability obeys $h_j \geq 1/2 - a \geq 3/8$ and its estimated value is at least $1/2$. The log error of one factor is at most $(8/3)a$ by the mean-value theorem. The two products thus contribute at most $(16/3)Na = v/6$ to the log error in (40).

Connect x_0 to x_D by changing their differing coordinates one at a time, in any fixed order; there are at most N steps. If step j changes coordinate i_j of x^{j-1}, the log weight ratio is exactly

$$\log \frac{W^u(x_D)}{W^u(x_0)} = \sum_j (1 - 2x_{i_j}^{j-1}) \left[\ell_{i_j}(x_{\partial i_j}^{j-1}) + u_{i_j}\right]. \tag{41}$$

These are full local conditional log odds, not marginals under the rare pin event. Estimate each bracket to accuracy $v/(4N)$ with Lemma 5, including the actual u calibration and numerical evaluation in that tolerance. Allocate failure $\beta_0/(2N)$ per term. Their total log error is at most $v/4$ on a joint event of failure at most $\beta_0/2$. The chosen contexts are adaptive, but the conditional guarantees are uniform, so the union bound remains valid.

Compute a finite positive rational exponential of the resulting sum to log-relative error at most $v/12$, and multiply by the rational empirical heavy products. The total log error is at most $v/6 + v/4 + v/12 = v/2 < v$, leaving slack. All selected empirical factors are at least $1/2$. The absolute magnitude of (41) is at most $N(B_0 + U)$, so the rational output and computations have bit length polynomial in $N(B_0 + U)$, $\log(1/v)$, and the sample-count logarithms. There are at most $2N$ heavy stages, each using $M = O(N^2 v^{-2} \log(N/\beta_0))$ pinned draws. At most N balanced logit estimates have the same order of individual readings. This proves the claimed counts. □

One explicit cost form, with the constants of the preceding proof, is

$$\begin{aligned} C_{\mathrm{mass}}(v, \beta_0) \;&\leq 2NM\, C_{\mathrm{pinlaw}}(v/(64N)) \\ &\quad + N\, C_{\mathrm{odds}}(v/(4N), \beta_0/(2N)) + C_{\mathrm{ratio-arith}}, \quad M = \left\lceil 2(32N/v)^2 \log(8N/\beta_0) \right\rceil. \end{aligned} \tag{42}$$

Here C_{odds} includes every actual calibration, setting, slew, settling, hold, read and timer call in Lemma 5.

7.1 From pin masses to disconnection Count and SampleB

Let the physical failure event B be all-terminal disconnection of the declared graph, with bit 1 denoting an edge failure. The full failure rule is supplied. The underlying graph is connected and has at least two vertices, so B is nonempty and the cut weights have a positive minimum. The coarse rational bounds (35), with fixed c, yield rational cut weights as in the accepted correlated construction. For $C = \lceil 2c \rceil$, that construction has two branches. If its minimum cut weight λ_{cut} is below $4\lceil \log_2(2v_{\mathrm{graph}}) \rceil$, it supplies the explicit lower bound $p_0 = 2^{-8C\lceil \log_2(2v_{\mathrm{graph}}) \rceil}$ on $\pi^u(B)$; ordinary Gibbs rejection then costs at most $v_{\mathrm{graph}}^{O(C)}$. In the other branch, at target error h, it constructs at most

$$J_B \leq 64 v_{\mathrm{graph}}^{4C+6}/h$$

cut events and omits at most $(h/16)\pi^u(B)$ in mass on its declared enumeration event.

Use Lemma 8 to estimate each cut-pinning probability p_j by a positive rational ϕ_j with $e^{-s} p_j \leq \phi_j \leq e^s p_j$. Draw index j proportional to ϕ_j, then draw from the pinned law with Lemma 7. For exact component draws, the proposal density is

$$\nu(x) = \frac{\pi^u(x)}{\sum_j \phi_j} \sum_j \frac{\phi_j}{p_j} \mathbf{1}_{E_j}(x).$$

With $d(x) = \sum_j \mathbf{1}_{E_j}(x)$, the ratio of the sum in this expression to $d(x)$ lies in $[e^{-s}, e^s]$. The reciprocal statistic $1/d(x) \in [1/J_B, 1]$ therefore gives relative union counting by a bounded-mean estimator; accepting with probability $1/d(x)$ gives the union-conditioned law with a multiplicative density error $e^{\pm 2s}$. This is the exact place where relative pin mass, rather than knowledge of W, is needed.

For counting at relative error h, the accepted schedule uses $s = h/16$, joint pin-mass confidence error at most one quarter of the requested count failure probability, and at most

$$\lceil 2048 J_B h^{-2} \lceil \log_2(8/\beta_0) \rceil \rceil$$

union proposals. For a conditional draw at TV error h, it uses a fresh list, total enumeration/mass failure at most $h/8$, $s = h/32$, and at most $\lceil 2J_B \lceil \log_2(16/h) \rceil \rceil$ proposals. It always returns a state in B, using the known all-failed state as the finite-cap fallback. Finite categorical and reciprocal arithmetic are the accepted finite rational procedures. All requested relative pin accuracies and confidences are thus explicit functions of h, β_0, J_B.

Dependent physical calls are handled by conditional means, rather than requiring every sample to succeed on a shared event. Condition on the good retained-list and mass-estimate event. For the count statistic $Y = 1/d(X)$, every subsequent conditional mean differs from the ideal mixture mean by at most the proposal-TV and rounding budget $h/(64J_B)$. The ideal mean is at least $1/J_B$. Conditional bounded-variable exponential-moment concentration, proved in Lemma 10, therefore gives the same relative concentration schedule, with confidence entering only logarithmically. For SampleB, every proposal conditional on every previous history is within its allocated TV of the same ideal proposal. Its accepted mark law is uniformly close to the same union-conditioned law by Lemma 11. A mixture over the possible first accepted histories retains that bound, and the cap has the same exponential tail. Thus the accepted static tolerance schedule is valid with fresh conditional-on-history physical calls, without an iid assumption. Running it at half the requested error supplies slack for conservative constants. Pin-mass confidence failures for SampleB are included in its unconditional TV budget; no union over every future SampleB invocation is demanded.

Consequently the exact interfaces needed by the accepted first-event sampler are now physical finite operations:

$$\mathrm{Count}(h, \beta_0), \qquad \mathrm{SampleB}(h),$$

with conditional-on-history confidence/TV guarantees for the true law. A fresh SampleB call rebuilds its random estimates. Reusing a shared random table instead requires an explicitly allocated outer confidence event; it is not an independent sample by declaration.

7.2 Optional fitted-static alternative

A distinct implementation may use the accepted active learner to construct a rational static weight table $\widehat{W}$ satisfying, on one shared confidence event, $e^{-E} \le \widehat{W}(x)/W(x) \le e^E$ for every x. Then

$$e^{-2E} \le \frac{\widehat{\pi}(B)}{\pi(B)} \le e^{2E}, \qquad d_{\mathrm{TV}}\big(\widehat{\pi}(\cdot \mid B), \pi(\cdot \mid B)\big) \le \tanh E.$$

Indeed normalization on the whole space or on B changes a pointwise weight ratio by at most another factor e^E, and a likelihood ratio in $[e^{-2E}, e^{2E}]$ has TV at most $\tanh E$. Choose E to spend at most half the requested count/sample error; the known-model static construction supplies the other half. Its pinned mixing and fixed contrast must be certified for the fitted weights too. True physical $P = P^*$ updates still use the operating-exit factory, so no fine kinetic table is needed. This hybrid can amortize an acquired static table over many records. The on-demand weight-ratio route is an alternative, not a universal cost improvement over that hybrid.

8 Conditional errors, finite confidence and the complete operation

The local cells can retain arbitrary previous states. What the proofs supply is a response kernel uniformly close to the intended one, conditional on every preceding experimental transcript. They do not assert that reused physical cells are independent.

Lemma 9 (Finite adaptive substitution)**.** *If a finite controller makes at most D adaptive calls, and each conditional response differs from its ideal kernel by at most θ_j, with $\sum_{j \le D} \theta_j \le a$ deterministically, its whole transcript differs from the ideal one by at most a in TV.*

Proof. Couple the classical randomness. Before the first mismatch the requested inputs agree. Couple the next response maximally, and sum the conditional first-mismatch probabilities. The claim holds for predictable allocations and nested calls with finite caps. □

A naive choice to implement every *whole static sampler* at accuracy δ/D would unnecessarily worsen its accuracy cost. There are two valid alternatives: flatten to primitive physical coins whose accuracy cost is logarithmic, paying their actual metrology services, or use conditional-mean concentration directly. The latter gives a transparent confidence interface.

Lemma 10 (Bounded conditional-mean calibration)**.** *Let $Z_1, \dots, Z_n \in [0,1]$ obey, for every prior transcript, $\mathbb{E}[Z_j \mid \mathcal{F}_{j-1}] \in [\mu - b_0, \mu + b_0]$. Suppose $\mu \geq q_0 > 0$, $0 < \xi \leq 1/2$ and $b_0 \leq \xi q_0/16$. Then*

$$n \geq \frac{32}{\xi^2 q_0} \log \frac{2}{\delta} \quad \Longrightarrow \quad \mathbb{P}(|n^{-1} \sum_j Z_j - \mu| > \xi\mu) \leq \delta.$$

Proof. Convexity on $[0,1]$ gives $\mathbb{E}[e^{tZ_j} \mid \mathcal{F}_{j-1}] \leq \exp((\mu + b_0)(e^t - 1))$ for $t > 0$. For $t < 0$ the same bound uses $\mu - b_0$. Iterating conditional expectations gives the usual Bernoulli upper and lower exponential bounds with these two fixed means; independence is not needed. The distance between either target threshold $\mu(1 \pm \xi)$ and the appropriate comparison mean is at least $15\xi\mu/16$. The elementary multiplicative Chernoff inequalities then bound the sum of the two tails by $2e^{-nq_0\xi^2/12}$; the displayed constant is conservative. □

Lemma 11 (Conditional accepted records)**.** *An ideal attempt outputs rejection or an accepted mark, with success probability $q \geq q_0 > 0$ and conditional mark law ν. Suppose that after every preceding history the actual combined rejection/mark law is within TV $a \leq q_0/2$ of this ideal law. Its conditional successful mark is within $2a/q_0$ of ν. Capping at J attempts and returning an arbitrary fallback mark on exhaustion has output TV at most $2a/q_0 + e^{-q_0 J/2}$ from ν.*

Proof. The success mass changes by at most a. Normalizing the two accepted submeasures and adding and subtracting their common-denominator form gives the stated $2a/q_0$ bound. This bound is uniform over every pre-success history and hence over their mixture. Each conditional success probability is at least $q_0/2$, so exhaustion has probability at most $(1 - q_0/2)^J \leq e^{-q_0 J/2}$. No independence is assumed. □

For Palm calibration, Z is the rounded reciprocal visit count. A proposal's conditional TV error bounds its conditional mean bias, and the accepted floor is $q_0 = g/8$. Choose this bias at most $\xi q_0/16$; the number of proposals remains $O(g^{-1}\xi^{-2}\log(1/\delta))$. Count's reciprocal-overlap statistics similarly have a polynomial floor $1/J_B$. Heavy-prefix marginal estimation already uses martingale Hoeffding. These facts preserve logarithmic confidence dependence in the abstract query counts. An apparatus service such as $C_{\text{field}}(\theta)$ may have a much worse precision dependence; its physical cost is still paid.

Theorem 2 (Acquired joint-event generation from local queries)**.** *Under the faithful-cell model, calibrated and reproducible environments, finite physical services, certified pinned margin and fixed contrast above, the all-terminal disconnection first-event record admits the following finite operation. With prescribed shared confidence $1 - \delta$, a compiled generator's next record, conditional on every previous experimental transcript, is within prescribed TV error ε of the true quantized record from a supplied initial law. The initial-law approximation is included in the error budget. The generator uses fresh local experiments on every draw, without a fine table of operating rates. A fitted static table is optional.*

Compilation and per-record query counts are finite and polynomial in the displayed static interfaces at fixed contrast exponent, numerical inverse mixing margins, $\log(1/\zeta)$, the value of the requested window, clock digits, inverse error and log inverse confidence. Degree dependence is that of the chosen coarse certificate and the priced neighborhood interface. There is no inverse global-rarity or mean-lifetime trajectory factor. Every apparatus service is evaluated at its actual requested accuracy and its cost is added.

Proof. The three preceding constructions supply Count, SampleB and $P = P^*$ for one and the same true invariant law and operating generator. Their conditional errors can be budgeted through finite adaptive substitution for records and through the conditional concentration lemma for calibration. The accepted first-event compiler then applies, including its rare/direct branch, arbitrary-start correction, finite clock, numerical guards and capped loops. Its mathematical Palm and mixing arguments do not require the unknown physical rates to be rational; the physical kernel calls replace their finite numerical evaluation. Allocate, for example, half the final record error to its finite construction and the remainder to physical substitutions and initialization, with the corresponding independent confidence budgets for structural and

static/shared calibration events. Each nested budget has a finite call cap and can be refined before execution. All quantities are common to the true physical law; no proxy stationary law is inserted without its own error bound. □

For explicit accounting, let $M_{a,G}, M_{a,P}, M_{a,O}$ be the finite numbers of Gibbs updates, operating updates and balanced log-odds estimates after expanding the accepted static/sampler schedules, with $a = 0$ for compilation and $a = 1$ for a record. Then

$$C_a \le C_{a,\mathrm{classical}} + \sum_{j=1}^{M_{a,G}} C_G(\theta_{a,j}) + \sum_{j=1}^{M_{a,P}} C_P(\theta_{a,j}) + \sum_{j=1}^{M_{a,O}} C_{\mathrm{odds}}(h_{a,j}, \beta_{a,j}). \tag{43}$$

Coarse acquisition, shared metrology, available hardware, propagation, stored configurations/cut lists/counters and chosen-intervention deployment/preparation are additional explicit entries. Parallelism changes wall time only by paying the corresponding cells and services. For K menu items and n records per loss estimate, the decision pays the K compilations and Kn per-record costs. Conditional Hoeffding and the same empirical-minimizer argument as (33) give the bounded-loss decision guarantee without an independence assumption on actual reused cells.

For physical equilibrium-safe missions, keep the certified safe-mass and finite preparation conditions of Section 3; include preparation error and abort loss. With true experimental static draws, there is no fitted-weight $\tanh(E_u)$ term. The actual chosen operating environment must be reproduced. Approximate field reproducibility alone does not control an arbitrarily discontinuous unknown clock response. Either faithful reproduction is supplied, or a separate rate/law bound must be acquired and charged. This is one of the physical premises that the experiments do not eliminate.

9 What the representation changes

9.1 An explicit failure of full-lifetime information accounting

Let X be a two-state component, initially safe, with safe-to-failed rate λ and repair rate one. Independently let Y be a symmetric two-state chain with switch rate c, initially uniform. The event is the first failure of X. Compare it with a model having the same X and switch rate $\widehat{c}$ for Y. Both event CDFs are identical and both terminal Y laws are uniform.

Write $j = c\log(c/\widehat{c}) - c + \widehat{c}$. The KL divergence of the entire Y trajectory up to the first failure is j/λ. For a retained physical window h, however, condition on τ and use stationarity of Y: the retained segment has length at most h, common uniform initial law and KL at most hj. Thus the joint time, terminal state and window obey TV at most $\sqrt{hj/2}$, independently of λ. Long stretches of unreported ordinary behavior can carry arbitrarily much model-discriminating information while contributing no corresponding uncertainty to the retained first-event question.

This observation alone is not an algorithm for interacting systems. The proposal construction in Section 2 supplies the computable interacting-system replacement: local divergences are paid on a short forward/reverse proposal, its successful record is controlled through accepted submeasures, and only a relative block-frequency error is paid by the binary waiting clock.

9.2 Comparison with strong applicable methods

Goal-oriented path-space uncertainty is established machinery. Birrell, Katsoulakis and Rey-Bellet explicitly treat stopping-time distributions and general stopped observables using appropriate relative-entropy objects [2]. Rényi-divergence methods give rare-event sensitivity bounds that can remain useful where ordinary event-probability estimates fail [3]. The contribution here is the explicit local-evidence-to-marked-proposal bound, the nonrare conditioning floor, and its finite acquisition/generation/decision composition. We do not claim that all preceding uncertainty methods pay inverse rarity, or that the use of an information distance is new.

Reverse-time sequential Monte Carlo already targets stopped rare paths [8]. Blanchet obtains strongly efficient overflow-path sampling for Jackson networks, with work proportional to the path length in the stated network family [9]. Parallel replica methods use quasi-stationary structure to accelerate metastable trajectories [10]. These are substantial existing operations, and exact finite-state solvers can be superior for small systems. The present construction adds a finite joint-TV and experimental-interface certificate for its declared succinct interacting model class; it is not a universal replacement for those algorithms. Our certified gap and static-access premises can fail in systems where a specialized rare-path method works.

Bernoulli factories and sampling from unknown probability coins are also established [4, 5]. Our release race and bounded random-walk factory are explicit elementary realizations for the particular physical rate interface, not claims to invent the factory principle. Active perturbations and nonequilibrium data can already make spin-model inference much easier [11, 12]. The new operational endpoint joins controlled local evidence to a finite rare first-event record and its intervention risk, while pricing actual clocks, fields, pins, preparation and fresh-query use.

A targeted search also found 2026 primary abstracts by Sun, Moody and Barthel (arXiv:2608.16171) and Barthel, Sun and Peng (arXiv:2608.17511) describing conditional-path Monte Carlo with rejection-free spacetime cluster updates. Their full texts could not be retrieved in this session. We record this relevant newer alternative without asserting a theorem-level separation from it or using its claims as proof premises.

10 Executed evidence and its limits

10.1 An exact-rational uncertainty calculator

The accompanying `certificate.py` calculates conservative upper bounds with rational arithmetic. For $0 \le a < 1$, it replaces $e^a - 1$ by $a/(1-a)$; it bounds e^{-gb}/ζ by $2^{-\lfloor gb \rfloor}/\zeta$ and rounds square roots upward using integer square roots with verified rational endpoints. For $0 \le d < 1$, it bounds $e^d - 1$ by $d/(1-d)$. Exact rational certificates are saved separately from decimal display values. This is not a general Gibbs counting or first-event simulation engine.

The three-state example of Section 1 was executed with $r = 0.4$, $\widehat{r} = r(1 + 10^{-12})$ and $\widehat{\lambda} = \lambda(1 + 10^{-12})$. The weights $(1, \lambda r, \lambda(1-r))$ and their counterparts satisfy $E = a = 2 \cdot 10^{-12}$. The common choices $R = 4$, $g = 1/4$, $\zeta = \lambda/8$, $p_+ = 2\lambda$ are valid. For $H = 1$ and schedule parameter $\varepsilon = 0.2$, the calculation gave:

Mean first time	Buffer b	Proposal length L	Certified model TV
10^{12}	212	4,403,200	< 0.004909
10^{30}	448	9,236,480	< 0.006221
10^{100}	1,380	28,323,840	< 0.005368

In this example the pre-event state is constant, so the precursor record is determined by the time and terminal mark. An independent analytic density integral gives approximately $4.8405290 \cdot 10^{-13}$ for the actual time-and-mark TV, the same for all three λ values. That value is a high-precision floating diagnostic, not the rational certificate. The gap between it and the certificate shows how loose the generic sufficient schedule is. The ordinary exact three-state solver is far better for this example; this table establishes neither speedup nor practical acquisition cost. It checks the new bound and its absence of a numerical lifetime factor. The general ordered-precursor assertion rests on the proof, not this trivial precursor execution.

An initial attempted $\lambda = 10^{-6}$ instance failed the calculator's rare-branch condition because its prescribed block was too long for $Lp \le 1$. The guard rejected it; no certificate was issued. The common branch theorem instead directs such inputs to the finite direct branch. The diagnostic subsequently used only the displayed rare instances. This failed attempt is retained as a scope check.

10.2 Independent constructions retained

The separate decision-route execution checks its finite local factories and static weight-ratio identity: 10 exact rational bounded-walk cases, 20 balanced-Gibbs factory cases, and 4 independently enumerated pin-mass identities passed. The cases include operating rate 2^{-100} and a pin event of probability approximately $1.45 \cdot 10^{-11}$. These are exact arithmetic checks of finite formulas, not observations of those rare outcomes in a physical system.

The service example is explicit. For a single operating update at $\theta = 2^{-20}$ and $\rho = 2$, the prescribed cap is 1,656 pins, a deadline of 19.5 model time units per race (20 if rounded to whole units), and aggregate capped dwell 32,292 (33,120 when rounded). The comparison timing allowance is at most $1.406 \cdot 10^{-13}$ model time units, with a sufficient 109-bit deadline-uniform draw. Context setup, pins, settling, detector service, reference-clock metrology and computation are added. This substantial bill is sufficient, not necessary, and demonstrates no economical device. A full event forecast includes many such calls and the static operations.

The complete record states which primitives were simulated and charges the finite call caps. No physical local cell, detector or controlled network was built. Neither that execution nor the present calculator implements the full correlated static cut algorithm on a large network.

An independent foundational route produced an explicit finite rational compiler from two-counter programs to globally certified 1/2-contractions of $\mathbb{R}^4$ with known unique fixed point zero. Exact finite arrival at zero and one fixed homogeneous sign event encode halting. Its finite compiler and exact arithmetic checks were executed. The late distinguishing signal shrinks exponentially; positive absolute event margins have a finite logarithmic horizon. General piecewise-affine reachability undecidability is prior machinery and scaling may already imply this consequence. It is retained as an exact-versus-finite-resolution boundary, without a priority or robust physical-computation claim. Its full proof, primary comparisons and precision bill are in `next5_fixedlaw`.

The independent mechanical route asks whether active forces can replace an absent relaxation certificate when inferring zero-frequency transport. For its equilibrium linear family, true transport is $K = M$ versus reference $K = 1$, while every causal force controller with $\int_0^T u_t^2 dt \le E$ satisfies

$$D_{\mathrm{KL}}(P_{a,M}^{\mathrm{record}} \| P_0^{\mathrm{record}}) \le a(1+T) + a^2 E T^2, \quad 0 < a \le 1/4, \quad M \ge 2.$$

Sending the visible coupling a toward zero hides arbitrarily large transport from any fixed finite time/effort budget. Its separate finite-slew, specified-pair pulse reaches the cubic active-visibility scale $U^2 a^2 T^3$. The proof and finite-meter/work-reserve diagnostic retain their supplied equilibrium initialization, force port and exact model-order qualification. The reference has minimal order one; the alternatives have order two. No fixed-reference stopping impossibility is asserted for an independently certified exact-order-two class. These results do not alter the declared-model premises of the acquired-event construction.

11 Preservation and completion ledger

The complete preceding first-event package contains 182 files, including its preserved predecessors. The new package retains those originals byte for byte, with the baseline and final SHA-256 comparison. Earlier quantum acquisition/control/calibration, effusion, reliability, transport, passive-memory, fixed-law, active acquisition/prevention and fixed-mechanics results keep their original hypotheses and records.

The new completed objects are the marked-law comparison and its local envelopes; the common finite branch and acquired uncertainty/decision composition; the experiment-fed static and dynamic interfaces with their finite caps; and the two independent boundary constructions. The package also keeps exact source files, programs, execution outputs, resource accounting, primary-source links, and unsuccessful approaches. No original proof is replaced by a relaxed summary.

The substantive outstanding engineering task is economical realization in a specified physical system: faithful cells, calibrated strong fields, low-cost pins, fast detectors and strict mixing margins must actually be available. The theorem does not claim that such hardware or an economical general implementation has been demonstrated. It resolves the declared scientific question: local controlled access can support a joint rare-event forecast and a quantitative permitted intervention decision without observing a global failure or simulating its full lifetime, with all remaining access and numerical costs exposed.

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